Calculus Lifesaver

Contents

Chapter 10: Infinite Sequences and Series

Theorem 1: Sequence limit calculations

Theorem 2: The Sandwich Theorem for Sequences

Theorem 3: The Continuous Function Theorem for Sequences

//Treat sequences as a function with $\mathbb{N}^+$ as its domain

Theorem 4: From the limit of a function to infer the limit of a sequence

Definitions: Bounded, Least Upper Bound, Greatest Lower Bound, Nondecreasing/Nonincreasing, Monotonicity,

Theorem 5: Commonly Occurring Limits

Theorem 6: The Monotonic Sequence Theorem

Definitions: nth partial sum, Infinite series ($a_1 + a_2 + \cdots + a_n + \cdots$), the sequence of the partial sums, ***the sum of infinite series*** (which doesn't exist but is defined *for convenience*)

Definition: Geometric Series ($a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}$ with a, r fixed)

Theorem 7: The nth-Term Test for Divergence

Theorem 8: Algebraic Operations for the Limit of the Sum of Convergent Series

Corollary: Addition/Subtraction on two series that one is convergent and one is divergent produces a divergent series.

Theorem 9: The Integral Test (**Positive**, decreasing, continuous function)

Definition: ***p-series***: when $p = 1$, it is the **harmonic series**.

Theorem*: Bounds for the Remainder in the Integral Test (Estimating the Error)

$$\int_{n+1}^{\infty} f(x)dx \leq R_n \leq \int_{n}^{\infty} f(x)dx$$

Corollary*: $s_n + \int_{n+1}^{\infty} f(x)dx \leq S \leq s_n + \int_{n}^{\infty} f(x)dx$

Theorem 10: The Comparison Test (Nonnegative terms)

Theorem 11: Limit Comparison Test (Nonnegative terms)

Example: Convergent *p*-series

Definition: Absolute Convergence

Theorem 12: The Absolute Convergence Test

Caution: Convergence does not **imply** absolute convergence.

Theorem 13: The Ratio Test (effective for a series contain factorials or powers)

Theorem 14: The Root Test

Theorem 15: The Alternating Series Test (Positive; Eventually Nonincreasing; $u_n \to 0$)

Example: Differentiating w.r.t n to show that a sequence is nonincreasing

Theorem 16: **The Alternating Series Estimation Theorem** ($|\epsilon| < u_{n+1}$, the first unused term)

Definition: Conditional Convergence

Theorem 17: The Rearrangement Theorem for Absolutely Convergent Series

Definition: Power Series (Center, Coefficients)

Theorem 18: The Convergence Theorem for Power Series (absolutely converges for $|x| < |c|$ and diverges (doesn't converge even conditionally) for $|x| > |c|$)

Definition: The Radius of Convergence of a Power Series

Corollary to Theorem 18: (Radius of Convergence; For all x convergent; Converge only at $x = 0$)

Theorem 19: The Series Multiplication Theorem for Power Series

Theorem 20: Substitution for Convergent Power Series

Theorem 21: The Term-by-Term Differentiation Theorem (for power series)

Theorem 22: The Term-by-Term Integration Theorem

$$\frac{\pi}{4} = \arctan 1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$$

$$\ln 2 = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}x^n}{n}$$

Definition: Taylor Series; Maclaurin Series (Taylor series at $x = 0$)

Definition: Taylor Polynomials

Theorem 23: Taylor's Theorem (f and its first n derivatives are continuous on the closed interval between a and b, and $f^{(n)}$ is differentiable on the open interval between a and b, then there exists…)

Equation: Taylor's Formula

$$f(x) = P_n(x) + R_n(x)$$

Theorem 24: The Remainder Estimation Theorem

$$|R_n(x)| \leq M \frac{|x-a|^{n+1}}{(n+1)!}$$

Definition: Binomial Series (Converges for

$$(1+x)^m = \sum_{k=0}^{\infty} \binom{m}{k} x^k$$

Chapter 11: Parametric Equations and Polar Coordinates

Definition: Parametric Curve, Parametric Equations, Parametrization

Definition: **Natural Parametrization**

Definition: Cycloid, Brachistochrone (shortest-time curve)**, Tautochrone** (same-time curve)

$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$$

Theorem*: Parametric Formula for dy/dx(If all three derivatives exist and $\frac{dx}{dt} \neq 0$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Theorem*: Parametric Formula for dy'/dx

Definition: **Astroid**

Definition: Smooth Curve: $|v(t)| \neq 0$

Definition: Arc Length Differential

$$ds = \sqrt{[f'(t)]^2 + [g'(t)]^2} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

Theorem*: Revolution about an axis (for example, x-axis)

$$S = \int_a^b 2\pi y \, ds$$

Definition: Polar Coordinates

Definition: **Lemniscate**

Theorem*: Area of the Fan-Shaped Region Between the Origin and the Curve $r = f(\theta)$ where $\alpha \leq \theta \leq \beta$

$$A = \int_\alpha^\beta \frac{1}{2} r^2 \, d\theta$$

Where the area differential is defined to be

$$dA = \frac{1}{2} r^2 d\theta = \frac{1}{2}\big(f(\theta)\big)^2 d\theta$$

Theorem*: Length of a Polar Curve (Traversing exactly **once**)

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

Chapter 12: Vectors and the Geometry of Space

Definition: Cartesian Coordinates, Distances (Measure Theory), Standard Equation for a Sphere

Definition: **Vector**, Initial Point, Terminal Point, Length, Component Form

$$\langle v_1, v_2 \rangle$$

Definition: **Vector Space**, Addition, Scalar Multiplication

Definition: **Unit Vector,** Midpoint

$$\hat{v} = \frac{v}{|v|}$$

Definition: **Dot Product**

$$\mathbf{u} \cdot \mathbf{v} = \sum_{k=1}^{n} u_k v_k$$

Theorem 1: Angle Between Two Vectors

$$\theta = \arccos\left(\frac{\sum_{k=1}^{n} u_k v_k}{|\mathbf{u}||\mathbf{v}|}\right) = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}\right)$$

Definition: Projection

$$\text{proj}_{\text{v}}\, \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\right) \mathbf{v}$$

Definition: Work

$$W = \int_i^j \mathbf{F} \cdot ds$$

Definition: Cross Product

$$\mathbf{u} \times \mathbf{v} = (|\mathbf{u}||\mathbf{v}| \sin \theta)\mathbf{n}$$

Theorem*: Nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$

Properties of the Cross Product

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$$

Definition: Triple Scalar Product (Volume of a Parallelepiped Box)

$$|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| = |\mathbf{u} \times \mathbf{v}||\mathbf{w}| \cos \theta$$

Properties of The Triple Scalar Product

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = (\mathbf{v} \times \mathbf{w}) \cdot \mathbf{u} = (\mathbf{w} \times \mathbf{u}) \cdot \mathbf{v}$$

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$$

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

Theorem*: Vector Equation for a Line

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$$

Theorem*: Distance from a Point S to a Line Through P Parallel to $\mathbf{v}$

$$d = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$$

Theorem*: Equation for a Plane

$$\mathbf{n} \cdot \overrightarrow{P_0P} = 0$$

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

$$Ax + By + Cz = D, \text{ where } D = Ax_0 + By_0 + Cz_0$$

Theorem*: Two planes are parallel IFF their normal are parallel.

Theorem*: Distance from a Point S to a Plane

$$d = \left|\overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|}\right|, \text{ where } \mathbf{n} \text{ is normal the plane}$$

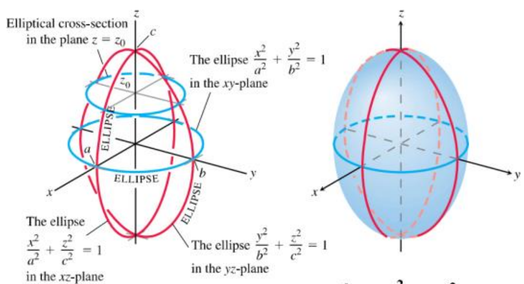
Theorem*: Angle Between Planes: Angle between their **normal.**

Definition: **Cylinders, Generating Curve**

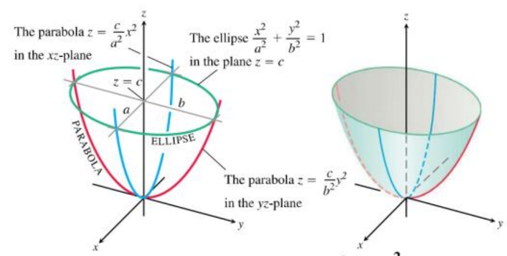
Definition: Quadric Surfaces

$$Ax^2 + By^2 + Cz^2 + Dz = E$$

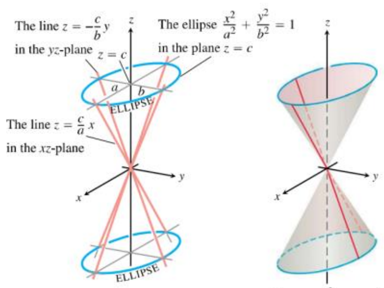
TABLE 12.1 Graphs of Quadric Surfaces



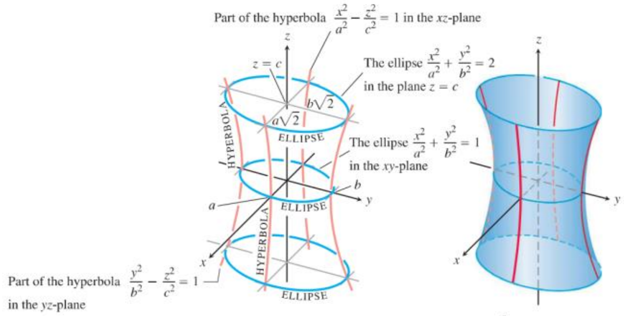
ELLIPSOID $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$



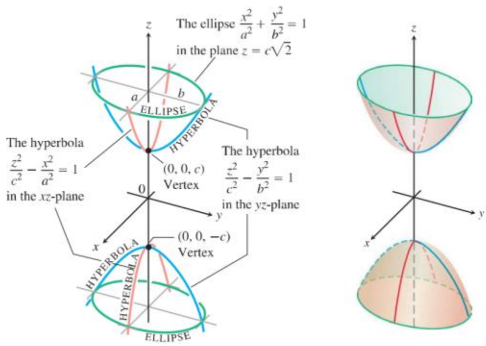
ELLIPTICAL PARABOLOID $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$



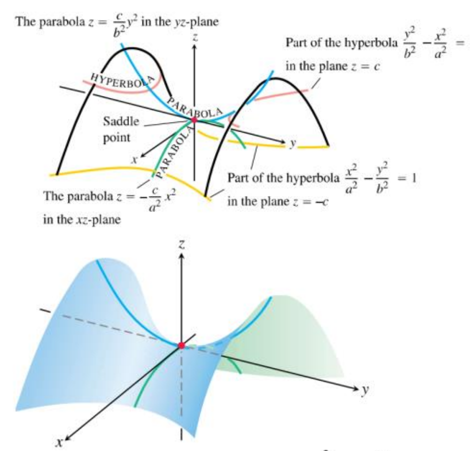
ELLIPTICAL CONE $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$



HYPERBOLOID OF ONE SHEET $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$



HYPERBOLOID OF TWO SHEETS $\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$



HYPERBOLIC PARABOLOID $\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}, c > 0$

Definition: Helix

Chapter 13: Vector-Valued Functions and Motion in Space

Definition: **Limit, Continuous**: $\lim_{t \to t_0} \mathbf{r}(t) = \mathbf{r}(t_0)$

Definition: Derivative (**is differentiable**) at t

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{df}{dt}\mathbf{i} + \frac{dg}{dt}\mathbf{j} + \frac{dh}{dt}\mathbf{k}$$

Definition: **Smooth** (derivative is continuous and never 0)

$$\frac{d\mathbf{r}}{dt} \neq 0$$

Definition: Tangent Line (through P_0, parallel to $\mathbf{r}'(t_0)$)

Definition: **Piecewise Smooth**: Composed of Finite number of Smooth Curves

Definition: Velocity Vector, Speed, Accleration ector

Theorem*: **Differentiation Rules for Vector Functions**

$$\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$$

$$\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$$

Theorem*: Vector Function of Constant Length

$$\mathbf{r} \cdot \frac{d\mathbf{r}}{dt} = 0$$

Definition: Indefinite Integral of $\mathbf{r}$

$$\int \mathbf{r}(t)\, dt = \mathbf{R}(t) + \mathbf{C}$$

Definition: Length

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt = \int_a^b |\mathbf{v}|\, dt$$

Definition: **Unit Tangent Vector**

$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{d\mathbf{r}}{ds}$$

$$|\mathbf{v} = \frac{ds}{dt}$$

Definition: **Curvature**

$$\kappa = \left|\frac{d\mathbf{T}}{ds}\right|$$

$$\kappa = \frac{1}{|\mathbf{v}|}\left|\frac{d\mathbf{T}}{dt}\right| = \frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3}$$

Definition: **Principal Unit Normal Vector**

$$\mathbf{N} = \frac{1}{\kappa}\frac{d\mathbf{T}}{ds}$$

$$\mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/\mathrm{dt}|}$$

Definition: **Osculating Circle** or **Circle of Curvature**

$$\text{Radius of Curvature} = \rho = \frac{1}{\kappa}$$

Chapter 14: Partial Derivatives

Definition: Real-valued Function f on D, **domain, range, dependent variable, independent variable, input variable, output variable**

Definition: Interior Point, Boundary Point (*every* n-sphere contains **both** point from R and point not from R)

Definition: Interior, Boundary, Open (Consists *entirely* of interior points), **Closed** (Contains all its boundary points)

Definition: Bounded (Lies inside a disk of *finite* radius), **Unbounded** (If not bounded)

Definition: Level Curve: The set of points in the plane where a function $f(x,y)$ has a constant value $f(x,y)=c$ is called a **level curve of f.**

Definition: Graph, Surface (IFF $z=f(x,y)$)

Definition: Contour Curve (Different from *level curve*) On the graph (rather than a *projection*)

Definition: Level Surface: $f(x,y,z)=c$

Definition: Limit: A function $f(x,y,z)$ approaches the **limit** L as $(x,y,z)\to(x_0,y_0,z_0)$ and write

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} f(x,y,z)=L$$

if, for every number $\epsilon>0$, there exists a corresponding number $\delta>0$ such that for all (x,y,z) in the domain of f,

$$|f(x,y,z)-L|<\epsilon \text{ whenever } 0<\sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2}<\delta$$

Caution: The points (x,y) that approach (x_0,y_0,z_0) are always taken to be in the domain of f.

Theorem 1: Algebraic Operations on Limits of Functions

Example: Evaluating a limit *along different paths approaching the point.*

Definition: Continuous (Limit exists, and equals to the value of the function there)

Theorem*: Having the same limit along all straight lines approaching (x_0,y_0) does not imply a limit exists at (x_0,y_0)

Theorem*: Continuity of Composites

Theorem*: Extreme Values of Continuous Functions *must* exist in a Closed Bounded domain.

Definition: Partial Derivative

$$\left.\frac{\partial f}{\partial x}\right|_{x_0,y_0} = \lim_{h\to 0}\frac{f(x_0+h,y_0)-f(x_0,y_0)}{h}$$

Theorem 2: The Mixed Derivative Theorem (Open region, all continuous at the point)

Definition: (*Differentiability* of a multi-variable function)

$$\Delta z = f_x(x_0,y_0)\Delta x + f_y(x_0,y_0)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta x$$

In which each of $\epsilon_1, \epsilon_2 \to 0$ as both $\Delta x, \Delta y \to 0$.

Theorem 3: The Increment Theorem for Functions of Two Variables (Defined, Open Region R containing the point, f_x and f_y are continuous at (x_0, y_0). Then f is differentiable) (*Existence and Continuity of Partial Derivatives implies Differentiability*)

Theorem 4: *Differentiability Implies Continuity*

Theorem 5: Chain Rule For Functions of One Independent Variable and Two Intermediate Variables

$$\frac{dw}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Theorem 6: Chain Rule for Functions of One Independent Variable and Three Intermediate Variables

Theorem 7: Chain Rule for Two Independent Variables and Three Intermediate Variables

Theorem 8: A Formula for Implicit Differentiation (Continuous, Open region, $\exists c$ as a constant such that $F(x_0, y_0, z_0) = c$

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

Definition: Directional Derivative

$$\left(\frac{df}{ds}\right)_{\mathbf{u},P_0} = \lim_{s\to 0}\frac{f(x_0+su_1, y_0+su_2)-f(x_0,y_0)}{s}r$$

Or

$$(D_{\mathbf{u}}f)_{P_0}$$

Definition: The Gradient Vector

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{z}$$

$$(D_{\mathbf{u}}f)_{P_0} = (\nabla f)_{P_0}\cdot\mathbf{u}$$

Corollary: At every point (x_0, y_0, z_0) in the domain of a differentiable function $f(x, y, z)$, the gradient of f is ***normal*** to the level curve through (x_0, y_0, z_0).

Corollary: Tangent Line to a Level Curve

Theorem*: Algebra Rules for Gradients (Same as Rules for Calculating Derivatives)

Theorem*: Derivative Along a Path

$$\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$$

Definition: Tangent Plane on a *level surface* (normal to $\nabla f|_{P_0}$)

Theorem*: Estimating the Change in f in a Direction $\mathbf{u}$.

$$df = \left(\nabla f|_{P_0} \cdot \mathbf{u}\right) ds$$

Definition: Standard Linear Approximation

$$f(x, y) \approx L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Theorem: Error in the Standard Linear Approximation

Assume M is any upper bound for the values of $|f_{xx}|$, $|f_{yy}|$ and $|f_{xy}|$ on R, then the error $E(x, y)$ incurred in replacing $f(x, y)$ on R by its linearization satisfies the inequality

$$|E(x, y)| \leq \frac{1}{2} M(|x - x_0| + |y - y_0|)^2$$

Error in the Standard Linear Approximation

$$|E| \leq \frac{1}{2} M \left(\sum_{k=1}^{n} |u_k - u_{k0}| \right)^2$$

Definition: Total Differential of *f*

$$df = f_x(x_0, y_0)\, dx + f_y(x_0, y_0)\, dy$$

$$df = \sum_{k=1}^{n} f_{u_k}(P_0)\, du_k$$

Definition: Local Maximum, Local Minimum (n-sphere centered at P_0)

Theorem 10: First Derivative Test for Local Extreme Values (All first derivatives exist at P_0, the local extrema, and they are 0)

Definition: Critical Point (All partial derivatives are 0 or where ***at least one*** of the partial derivatives does not exist)

Definition: Saddle Point (Every n-sphere centered has both a larger and a smaller value)

Theorem 11: Second Derivative Test for Local Extreme Values (On Critical Points)
(First & second partial derivatives are continuous)

i. f has a **local maximum** at (a, b) if $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b).
ii. f has a **local minimum** at (a, b) if $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b).
iii. f has a **saddle point** at (a, b) if $f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a, b).
iv. **the test is inconclusive** at (a, b) if $f_{xx}f_{yy} - f_{xy}^2 = 0$ at (a, b).

Example: Finding Absolute Maxima and Minima on closed bounded regions

1. List the interior points
2. List the boundary points
3. Querying the list

Theorem*: Taylor's Formula for $f(x, y)$ at the origin

$$f(x,y) = f(0,0) + \ xf_x + yf_y + \cdots + \frac{1}{n!}\left(x^n \frac{\partial^n f}{\partial x^n} + nx^{n-1}\frac{\partial^n f}{\partial x^{n-1}\partial y} + \cdots + y^n \frac{\partial^n f}{\partial y^n}\right) + R_n$$

Where

$$R_n = \frac{1}{(n+1)!}\left(x^{n+1}\frac{\partial^{n+1} f}{\partial x^{n+1}} + (n+1)x^n \frac{\partial^{n+1} f}{\partial x^n \partial y} + \cdots + y^{n+1}\frac{\partial^{n+1} f}{\partial y^{n+1}}\right)\Bigg|_{cx,cy}$$

And is evaluated at a point on the line segment joining the origin and (x, y). **The expression of the remainder can be used to calculate the error involved.** (*Take the upper bound of all the partial derivatives as M, and substitute M for calculation.*)

Theorem 12: The Orthogonal Gradient Theorem (Differentiable function, smooth curve (parametrization)) implies (If P_0 is a point on C where f has a local extremum relative to its value on C, then ∇f is orthogonal to C at P_0)

Definition: Lagrange Multipliers

$$\nabla f = \lambda \nabla g$$

Where $g(x, y, z) = 0$.

Or with two constraints

$$\nabla f = \lambda \nabla g_1 + \mu \nabla g_2$$

Where both g_1 and g_2 valuates 0.

Chapter 15: Multiple Integrals

Definition: Iterated Integral (Repeated Integral)

Theorem 1: Fubini's Theorem (Continuous Function)

$$\iint_R f(x,y)dA = \int_c^d \int_a^b f(x,y)\,dx\,dy = \int_a^b \int_c^d f(x,y)\,dy\,dx$$

Theorem 2: Fubini's Theorem (*Stronger*) (Continuous Boundaries)

$$\iint_R f(x,y)dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$$

$$\iint_R f(x,y)dA = \int_c^d \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$$

Properties of Integration:

1. Constant Multiple
2. Sum and Difference
3. Domination
4. Additivity

Definition: Area (Closed, Bounded)

$$A = \iint_R dA$$

Theorem*: **Average Value** of f over R

$$f_{\text{avg}} = \frac{1}{\text{area of } R} \iint_R f\,dA$$

Theorem*: Fubini's Theorem (*In Polar Form*)

$$\iint_R f(r,\theta)dA = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=g_1(\theta)}^{r=g_2(\theta)} f(r,\theta)r\,dr\,d\theta$$

Definition: Triple Integral of f over D (with **reasonably smooth** boundaries)

$$\lim_{n\to\infty} S_n = \iiint_D F(x,y,z)\,dV$$

Definition: Volume (Closed, Bounded)

$$V = \iiint_D dV$$

Theorem*: Average Value

Formulas:

Mass:

$$M = \iiint_D \delta \, dV$$

Moment about the coordinate plane

$$M_{yz} = \iiint_D x\delta \, dV$$

Center of Mass

$$\bar{x} = \frac{M_{yz}}{M}$$

Definition: Cylindrical Coordinates: ordered triples (r, θ, z) in which $r \geq 0$.

Definition: Spherical Coordinates: ordered triples (ρ, ϕ, θ) in which $\rho \geq 0, 0 \leq \phi \leq \pi, \theta$ is the angle from cylindrical coordinates.

$$\lim_{n\to\infty} S_n = \iiint_D f(\rho, \phi, \theta) \, dV = \iiint_D f(\rho, \phi, \theta) \, \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

Definition: Jacobian Determinant

$$J(u, v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial y}{\partial u}\frac{\partial x}{\partial v} = \frac{\partial(x, y)}{\partial(u, v)}$$

$$J(u, v, w) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial u} \end{vmatrix}$$

Example:

$$\int_1^2 \int_{\frac{1}{y}}^{y} \sqrt{\frac{y}{x}} e^{\sqrt{xy}} \, dx \, dy = \int_1^2 \int_1^{\frac{2}{u}} 2ue^u \, dv \, du$$

Theorem 3: Substitution for Double Integrals (Continuous Function, Preimage R & image G **one-to-one**, g and h having continuous first partial derivatives within the interior of G.

$$\iint_R f(x,y)\,dx\,dy = \iint_G f\big(g(u,v),h(u,v)\big)\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

Chapter 16: Integrals and Vector Fields

Definition: Line Integral

$$\int_C f(x, y, z)\, ds = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k, y_k, z_k)\, \Delta s_k = \int_a^b f\big(g(t), h(t), k(t)\big)\, |\mathbf{v}(t)|\, dt$$

Definition: Gradient Field

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$$

Definition: Line Integral of F (Continuous Components)

$$\int_C \mathbf{F}\cdot\mathbf{T}\, ds = \int_C \left(\mathbf{F}\cdot\frac{d\mathbf{r}}{\mathrm{ds}}\right) ds = \int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}\big(\mathbf{r}(t)\big)\cdot\frac{d\mathbf{r}}{dt}\, dt$$

Theorem*: Integration by Components

$$\int_C M(x, y, z)\, dx = \int_a^b M\big(g(t), h(t), k(t)\big)\, g'(t)\, dt$$

$$\int_C N(x, y, z)\, dy = \int_a^b N\big(g(t), h(t), k(t)\big)\, h'(t)\, dt$$

$$\int_C P(x, y, z)\, dz = \int_a^b P\big(g(t), h(t), k(t)\big)\, k'(t)\, dt$$

Notation

$$\int_C M(x, y, z)\, dx + \int_C N(x, y, z)\, dy + \int_C P(x, y, z)\, dz = \int_C M\, dx + N\, dy + P\, dz$$

Definition: Work done by a force on a curve

Definition: Flow Integrals:

$$\text{Flow} = \int_{\mathrm{C}} \mathbf{F}\cdot\mathbf{T}\, ds$$

If the curve C starts and ends at the same point (so that $A = B$), the flow is called the ***circulation*** around the curve.

Definition: Flux Integrals: (Continuous vector field)

$$\text{Flux of } \mathbf{F} \text{ across C} = \int_C \mathbf{F}\cdot\mathbf{n}\, ds$$

Where $\mathbf{n}$ is the outward unit normal vector on C.

Theorem*: When a curve C is traversed counterclockwise on the x-y plane, the outward unit normal vector is $\mathbf{T} \times \mathbf{k}$.

Theorem: **Flux** of **F** Across a Smooth Closed Plane Curve (**Counterclockwise**)

$$(\text{Flux of } \mathbf{F} = M\mathbf{i} + N\mathbf{j} \text{ across } C) = \oint_C M\,dy - N\,dx$$

Definition: **Path Independence**

F is conservative on $\boldsymbol{D}$, or $\int_C \mathbf{F} \cdot d\mathbf{r}$ is **path independent in $\boldsymbol{D}$**.

Definition: Potential Function

$$\mathbf{F} = \nabla f$$

Corollary: Circulation on a conservative vector field **must be zero**.

Definition: Piecewise Smooth Curve

Definition: Connected (Join two points by a *smooth* curve)

Definition: Simply Connected (Any loop in D can be contracted to a point in D without ever leaving D)

Theorem 1: Fundamental Theorem of Line Integrals (Open Connected) (Let f be a differentiable function with a continuous gradient vector $\mathbf{F} = \nabla f$ on a domain D containing C.)

$$\int_{\mathbf{C}} \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A)$$

Theorem 2: Conservative Fields are Gradient Fields (Open Connected) (Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ be a vector field whose components are continuous throughout an open connected region D in space. Then **F** is conservative if and only if **F** is a gradient field ∇f for a differentiable function f.)

Theorem 3: Loop Property of Conservative Fields: The following statements are equivalent.

1. $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ around every loop in D.
2. The field is conservative on D.

Example: Component Test for Conservative Fields (Open Simply Connected) (Continuous Components)

$$\mathbf{F} \text{ is conservative IFF } \frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \frac{\partial P}{\partial x} = \frac{\partial M}{\partial z} \text{ and } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

This is derived from the **Mixed Derivative Theorem**.

Example (Component Test *Failed*)

$$\mathbf{F} = \frac{-y}{x^2+y^2}\mathbf{i} + \frac{x}{x^2+y^2}\mathbf{j} + 0\mathbf{k}$$

Definition: Exact Differential Forms

A differential form is **exact** on a domain D in space if

$$Mdx + Ndy + Pdz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz = df$$

For some scalar function throughout D.

This property is equivalent to $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ being a conservative vector field.

Definition: Divergence (Circulation Density)

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

Definition: Divergence (Flux Density), or the **k**-th component of the **curl**, or $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$.

$$\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}$$

Theorem: Green's Theorem

Circulation-Curl or Tangential Form (Counterclockwise circulation = **Curl** integral)

$$\oint_C \mathbf{F} \cdot \mathbf{T}\, ds = \oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy$$

Flux-Divergence or Normal Form (Outward flux = **Divergence** integral)

$$\oint_C \mathbf{F} \cdot \mathbf{n}\, ds = \oint_C Mdy - Ndx = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}\right) dx\, dy$$

Area Formula

$$\text{Area of } R = \frac{1}{2}\oint_C xdy - ydx$$

Definition: Parametrization of Surfaces

$$\mathbf{r}(u,v) = f(u,v)\mathbf{i} + g(u,v)\mathbf{j} + h(u,v)\mathbf{k}$$

Definition: Smooth Parametrization of a Surface (Both partial derivatives continuous)

$$\mathbf{r}_u \times \mathbf{r}_v \neq 0 \text{ (On the interior of the parameter domain)}$$

Definition: Area of the Smooth Surface

$$A = \iint_R |\mathbf{r}_u \times \mathbf{r}_v|\, dA = \iint_R |\mathbf{r}_u \times \mathbf{r}_v|\, du\, dv$$

Surface Area Differential

$$d\sigma = |\mathbf{r}_u \times \mathbf{r}_v|\, du\, dv$$

Theorem*: Formula for the Surface Area of an Implicit Surface

The area of the surface $F(x, y, z) = c$ over a closed and bounded plane region R is

$$\text{Surface area} = \iint_R \frac{|\nabla F|}{|\nabla F \cdot \mathbf{p}|}\, dA$$

Where $\mathbf{p} = \mathbf{i}, \mathbf{j}$, or $\mathbf{k}$ is normal to R and $\nabla F \cdot \mathbf{p} \neq 0$.

Formula for the Surface Area of a Graph

$$d\sigma = \sqrt{f_x^2 + f_y^2 + 1}\, dx\, dy$$

$$A = \iint_R \sqrt{f_x^2 + f_y^2 + 1}\, dx\, dy$$

Definition: Surface Integrals

$$\iint_S G(x, y, z)\, d\sigma = \lim_{n \to \infty} \sum_{k=1}^{n} G(x_k, y_k, z_k)\, \Delta\sigma_k$$

Definition: Orientation of a Surface

When we specify which of these normals we are going to use on consistently across the entire surface, the surface is given an *orientation*. A smooth surface is **orientable** (or **two-sided**) if it is possible to define a field of unit normal vector $\mathbf{n}$ on S which ***varies continuously*** with position. By convention, this $\mathbf{n}$ is usually chosen to ***point outward.***

Once $\mathbf{n}$ is chosen, we have **oriented** the surface, and call the surface together with its normal **oriented surface**. The vector $\mathbf{n}$ at any point is called the positive direction at that point.

Definition: Surface Integral of F over *S* is

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, d\sigma$$

Theorem*: Flux Across a Parametrized Surface (Assume $\mathbf{n} = \widehat{\mathbf{r}_u \times \mathbf{r}_v}$)

$$\text{Flux} = \iint_R \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v)\, du\, dv$$

Theorem: **Stokes' Theorem** (Piecewise Smooth, Oriented Surface) (Piecewise Smooth Boundary Curve C) (Continuous First Partial Derivatives, $\mathbf{F}$)

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$$

$$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$$

$$\oint_C \mathbf{F} \cdot \mathrm{d}\boldsymbol{r} = \iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} \, d\sigma$$

If two different oriented surfaces S_1 and S_2 has the same boundary C, their curl integrals are equal.

Theorem*:

$$\text{curl grad f} = 0$$

$$\nabla \times \nabla f = 0$$

Theorem 7: Curl $\mathbf{F} = \mathbf{0}$ Related to the Closed-Loop Property (Simply Connected) (Piecewise-Smooth Closed Path)

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$$

Theorem: Divergence Theorem: (Continuous First Partial Derivatives $\mathbf{F}$) (Piecewise Smooth Oriented Closed Surface)

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_D \nabla \cdot \mathbf{F} \, dV$$

Theorem 9: div $\mathbf{V} = \mathbf{0}$ if $\mathbf{V}$ = curl $\mathbf{F}$

$$\text{div (curl } \mathbf{F}) = \nabla \cdot (\nabla \times \mathbf{F}) = 0.$$